AETHERIA CELESTIAL OBSERVER

Full-Stack Architecture & Engineering Guide • Apple UI/UX Design System

Production Grade Architecture • Senior Full Stack Web Developer & Systems Engineer

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1. Executive Summary & Design Vision

Aetheria is an Apple-grade, high-performance celestial sky observation web application that renders a mathematically accurate, real-time 360-degree view of the cosmos from any vantage point on Earth. Strictly adhering to Apple's Human Interface Guidelines (HIG), the application fuses deep midnight aesthetics with translucent glassmorphism (Cupertino material blur), razor-sharp high-DPI instrumentation, and an organic, relaxing celestial soundscape generated in real time using the Web Audio API.

Unlike casual sky apps that display randomized decorative dots, Aetheria is anchored in verified astronomical astrometry sourced from the Yale Bright Star Catalog (BSC5) and Hipparcos J2000 epoch data. Every star, planet, constellation segment, and lunar phase is mathematically resolved against observer coordinates (Latitude and Longitude) and temporal epoch (Local Sidereal Time).

2. Full-Stack System Architecture

The application employs a decoupled, low-latency client-server topology engineered for 60 FPS rendering and offline fault tolerance:

Tier	Components & Tech	Key Responsibilities
Frontend Presentation	HTML5 Canvas 2D, CSS3, SF Pro, Vanilla JS	360° Spherical projection, High-DPI tape compasses, Glass UI
Audio Engine	Web Audio API, BiquadFilter, LFO	432Hz ambient drone, resonant breathing filter, stardust chimes
Time Engine	Pointer Events API, UTC epoch math	Apple play bar, glued thumb scrubber, Cupertino date picker
Backend Services	FastAPI, Uvicorn, Python 3.14	Astronomical ephemerides, star catalog API, document delivery

3. Celestial Mechanics & Astronomical Mathematics

3.1 Julian Date Conversion

```
For any Gregorian UTC timestamp (Y, M, D, H, Min, Sec):
If M ≤ 2: Y = Y - 1, M = M + 12
A = floor(Y / 100), B = 2 - A + floor(A / 4)
JD = floor(365.25 × (Y + 4716)) + floor(30.6001 × (M + 1)) + D + B - 1524.5
```

3.2 Sidereal Time & Local Hour Angle

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GMST = 280.46061837 + 360.98564736629 × (JD - 2451545.0) + 0.000387933 × T² - T³ / 38710000.0
LST = (GMST + Longitude) mod 360°
Hour Angle H = (LST - Right Ascension) mod 360°
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3.3 Equatorial to Horizontal Coordinates (Alt/Az)

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sin(h) = sin(δ) × sin(ϕ) + cos(δ) × cos(ϕ) × cos(H)
Altitude h = arcsin(sin(h))
cos(Az) = [sin(δ) - sin(ϕ) × sin(h)] / [cos(ϕ) × cos(h)]
If sin(H) > 0: Azimuth = 360° - arccos(cos(Az)), else Azimuth = arccos(cos(Az))
```

3.4 3D Spherical Perspective Projection onto Screen Viewport

```
Let camera orientation be (Acam, hcam) and Field of View be FOV. For an object at (A, h):
ΔA = A - Acam
vx = cos(h) × sin(ΔA)
vy = sin(h) × cos(hcam) - cos(h) × sin(hcam) × cos(ΔA)
vz = sin(h) × sin(hcam) + cos(h) × cos(hcam) × cos(ΔA)
When vz > 0 (object in front of camera):
x = ScreenWidth / 2 + (vx / vz) × [ScreenWidth / (2 × tan(FOV / 2))]
y = ScreenHeight / 2 - (vy / vz) × [ScreenWidth / (2 × tan(FOV / 2))]
```

4. Frontend Engineering & Apple Design System

High-DPI Anti-Blur Canvas Rendering: Standard HTML canvas elements suffer severe blurriness on Apple Retina displays due to the mismatch between CSS pixels and hardware device pixels. Aetheria calculates window.devicePixelRatio on every resize event, scales canvas buffer dimensions (canvas.width = cssW * dpr), and normalizes the 2D transformation matrix (ctx.scale(dpr, dpr)). All instrument ticks and text are snapped with +0.5px sub-pixel alignment for razor sharpness.

Linear Azimuth (Top) and Altitude (Left) Tape Compasses:
• Top Linear Compass: Spans 360° horizontally with tick marks every 1° (fine), 5° (medium), and 10° (labeled). Cardinal headings (N, NE, E, SE, S, SW, W, NW) are rendered prominently, with North highlighted in Apple Orange (#ff9f0a). A live digital heading badge (e.g. '184° S') floats above the center indicator.
• Left Linear Altitude Compass: Vertical tape spanning -90° (Nadir) to +90° (Zenith), with a vibrant green marker at 0° HORIZON and a digital elevation readout.

Timeline Scrubber Stickiness (Zero-Lag Pointer Physics): To satisfy the strict user requirement ('when I slide the bar the circle moves first, make sure the circle sticks with the bar'), the scrubber utilizes Pointer Events API (pointerdown, pointermove, pointerup) combined with element.setPointerCapture(pointerId). CSS transitions are explicitly disabled on the thumb during dragging, locking its center directly to (clientX - rect.left) / rect.width. Calculations for hour, minute, second and celestial coordinate recalculation occur synchronously in the same frame.

Apple iOS & macOS Cupertino 3D Drumroll Date Picker with Haptics Feedback:
• 3D Cylindrical Perspective: Columns for Month, Day, Year, Hour, and Minute render along a 3D cylindrical arc with rotateX, Z-translation, and optical foreshortening.
• Scroll-To-Rewind Interaction: Rolling or scrolling down on the drumroll or directly over the play bar date readout moves earlier dates into the central selection lens, scrubbing backward in time.
• Dual-Tier Haptic Feedback: Combines physical device vibration via navigator.vibrate(8) for touch devices with a synthesized 12ms acoustic mechanical notch click detent via the Web Audio API on desktop and macOS.
• Dynamic Calendar Constraints: Automatically reconciles days per month and leap years (clamping February to 28 or 29 days dynamically).
• Quick Astronomical Presets: One-tap jumps to celestial highlights (Current Real Time, Midnight Tonight, Perseid

Meteor Peak, Summer & Winter Solstices).

Apple Iconography & Controls:
• White Gemini Logo: Elegant 4-pointed celestial spark / twin glyph representing constellations.
• Flat White Minimal Telescope Logo: Clean, modern silhouette activating 3.5× magnification and a circular mil-dot reticle.
• Asteroid / Meteorite Logo: Stylized faceted meteor glyph for minor planets and comets.
• Minimal Apple Sound Icon: SF Symbols style speaker with live animated sound waves pulsating when ambience is active.
• Vertical Right Action Dock: Borderless 30% translucent liquid glass buttons on the right edge with zero 3D bulge.

Ambient Relaxing Sound Synthesizer: Constructed with Web Audio API nodes: a 432Hz fundamental drone, dual detuned sine/triangle oscillators for binaural warmth, a 380Hz Moog-style biquad lowpass filter with slow 0.065Hz LFO modulation, and randomized pentatonic stardust chimes (864Hz - 1728Hz) with stereo spatial panning. Smooth exponential gain ramps prevent any clicks or pops.

5. Backend Engineering & Microservices

The backend is developed with FastAPI and Uvicorn, exposing high-speed RESTful endpoints:
• GET /api/ephemeris: Resolves Sun, Moon, and 5 planets (Mercury through Saturn) plus bright stars for given timestamp and location.
• GET /api/stars: Returns the immutable star and constellation geometry database.
• GET /api/time: Returns synchronized atomic UTC time and Julian Date.
• GET /api/download/pdf & /api/download/docx: Serves the architecture documentation.

6. Step-by-Step Build & Deployment Procedure

```
Step 1: Install Python prerequisites:
    pip3 install fastapi uvicorn reportlab python-docx pillow

Step 2: Initialize project directory structure:
    backend/ (astronomy_engine.py, star_catalog.py, app.py)
    frontend/ (index.html, css/apple-theme.css, css/compass.css, js/)
    docs/ (generate_docs.py, ASTRONOMY_APP_ARCHITECTURE_GUIDE.pdf/docx)

Step 3: Launch FastAPI server:
    python3 -m uvicorn backend.app:app --host 0.0.0.0 --port 8000 --reload

Step 4: Open web browser to http://localhost:8000
    Interact with 360° celestial sphere, linear compasses, audio synthesizer, and timeline
scrubber.
```